Vector Integration

1. If $\bar{F}=3xy\bar{i}-y^2\bar{j}$, evaluate $\int \bar{F}.d\bar{r}$ where C is the curve $y=2x^2$ in x-y plane from (0,0) to (1,2).

Given $\bar{F}=3xy\bar{i}-y^2\bar{j}$.

Let $\bar{r}=x\bar{i}+y\bar{j}$

$d\bar{r}=\bar{i}dx+\bar{j}dy$

$\bar{F}dr=3xydx-y^2dy$

on $y=2x^2 \Rightarrow dy=4xdx$

$\bar{F}.d\bar{r}=3x(2x^2)dx-(4x)(2x^2)^2dx$

$=(6x^3-16x^5)dx$

$x: 0$ to 1

$$\int \bar{F}.d\bar{r}=\int_0^1 (6x^3-16x^5)dx$$

$$=\left[\frac{6x^4}{4}-\frac{16x^6}{6}\right]_0^1$$

$$=\frac{6}{4}(1)-\frac{16}{6}(1)$$

$$=\frac{9-16}{6}=-\frac{7}{6}$$

$$\therefore \int_C \bar{F}.d\bar{r}=\frac{-7}{6}$$

2. Find the work done in moving a particle in the force field $\bar{F}=3x^2\bar{i}+(2xz-y)\bar{j}+z\bar{k}$ along the straight line from (0,0,0) to (2,1,3)

Given $\bar{F}=3x^2\bar{i}+(2xz-y)\bar{j}+z\bar{k}$

Let $\bar{r}=x\bar{i}+y\bar{j}+z\bar{k}$

$d\bar{r}=\bar{i}dx+\bar{j}dy+dz\bar{k}$

$\bar{F}.d\bar{r}=3x^2dx+(2xz-y)dy+zdz$

Let $O=(0,0,0)$ and $A=(2,1,3)$

Equation of OA is

$$\frac{x-0}{2-0}=\frac{y-0}{1-0}=\frac{z-0}{3-0}=t$$

$x=2t$, $y=t$, $z=3t$

$dx=2dt$, $dy=dt$, $dz=3dt$

$t: 0 \to 1$

$$\bar{F}.d\bar{r} = 3(2t)^2(2dt) + (2(2t)(3t)-t)dt + (3t)(3dt)$$

$$= (24t^2 + (12t^2 - t) + 9t)dt$$

$$\bar{F}.d\bar{r} = (36t^2 + 8t)dt$$

$$\int \bar{F}.d\bar{r} = \int_{t=0}^{1} (36t^2 + 8t)dt$$

$$= \left[\frac{36t^3}{3} + \frac{8t^2}{2}\right]_0^1$$

$$= (12t^3 + 4t^2)_0^1$$

$$= 12 + 4$$

$$\int \bar{F}.d\bar{r} = 16$$

3. the work done by the force field $\bar{F} = z\bar{i} + x\bar{j} + y\bar{k}$, when it moves a particle along the arc of the curve $\bar{r} = \cos t\,\bar{i} + \sin t\,\bar{j} - t\bar{k}$ from $t=0$ to $t=2\pi$

Given $\bar{F} = z\bar{i} + x\bar{j} + y\bar{k}$

$$\bar{r} = \cos t\,\bar{i} + \sin t\,\bar{j} + t\bar{k}$$

$$d\bar{r} = -\sin t\,dt\,\bar{i} + \cos t\,dt\,\bar{j} - dt\,\bar{k}$$

Let $x = \cos t$, $y = \sin t$ & $z = -t$

$$\bar{F} = (-t)\bar{i} + (\cos t)\bar{j} + (\sin t)\bar{k}$$

$$\bar{F}.d\bar{r} = t\sin t\,dt + \cos^2 t\,dt - \sin t\,dt$$

$$\text{work done} = \int \bar{F}.d\bar{r}$$

$$= \int_0^{2\pi} (t\sin t\,dt + \cos^2 t - \sin t)\,dt$$

$$= \left[t(-\cos t) - (1)(-\sin t)\right]_0^{2\pi} + \int_0^{2\pi} \left(\frac{1+\cos 2t}{2}\right)dt + (\cos t)_0^{2\pi}$$

$$= (\sin t - t\cos t)_0^{2\pi} + \int_0^{2\pi} \left(\frac{1+\cos 2t}{2}\right)dt + (\cos t)_0^{2\pi}$$

$$= \left[\sin t - t\cos t + \frac{t}{2} + \frac{\sin 2t}{4} + \cos t\right]_0^{2\pi}$$

$$= \left(0 - 2\pi + \pi + \frac{0}{4} + 1\right) - (0 - 0 + 0 + 0 + 1)$$

$$= -2\pi + \pi + 1 - 1$$

$$= -\pi$$

$\therefore$ work done $= \int \bar{F} \cdot d\bar{r} = -\pi$

4. Evaluate $\int_S \bar{F} \cdot \bar{n}\, ds$ where $\bar{F} = 12x^2y\,\bar{i} - 3yz\,\bar{j} + 2z\,\bar{k}$ and S is the portion of the plane $x+y+z=1$ included in the first octant.

Let $\phi = 12x^2y - 3yz + 2z$

Let $\phi = x + y + z - 1$

Let $\nabla\phi$ be the normal to the surface ϕ

$$\therefore \nabla\phi = \bar{i}\frac{\partial\phi}{\partial x} + \bar{j}\frac{\partial\phi}{\partial y} + \bar{k}\frac{\partial\phi}{\partial z}$$

$$= \bar{i} + \bar{j} + \bar{k}$$

$$\nabla\phi = \bar{i} + \bar{j} + \bar{k}$$

unit normal $\bar{n} = \frac{\bar{i} + \bar{j} + \bar{k}}{\sqrt{1^2 + 1^2 + 1^2}} + \frac{\bar{i} + \bar{j} + \bar{k}}{\sqrt{3}}$

Let R be the projection of S on x-y plane

$$\int_S \bar{F} \cdot \bar{n}\, ds = \iint_R \bar{F} \cdot \bar{n}\, \frac{dx\,dy}{|\bar{n} \cdot \bar{k}|} \quad \text{---} (1)$$

$$\bar{F} \cdot \bar{n} = \frac{12x^2y - 3yz + 2z}{\sqrt{3}}$$

$$\bar{n} \cdot \bar{k} = \frac{(\bar{i} + \bar{j} + \bar{k})}{\sqrt{3}}(\bar{k})$$

$$\bar{n} \cdot \bar{k} = \frac{1}{\sqrt{3}}$$

from (1)

$$\int_S \bar{F} \cdot \bar{n}\, ds = \iint_R \frac{1}{\sqrt{3}}(12x^2y - 3yz + 2z)\frac{dx\,dy}{1/\sqrt{3}}$$

$$= \iint_R (12x^2y - 3yz + 2z)\, dx\, dy$$

Since R is the projection of S in xy plane, $z=0$

$x+y=1 \quad (z=0)$ $\qquad$ $x=1 \quad (y=0, z=0)$

$y=1-x$ $\qquad$ $x: 0$ to 1

$y: 0$ to $1-x$

$$= \int_{x=0}^{1}\int_{y=0}^{1-x} 12x^2y - 3y(1-x-y) + 2(1-x-y)\, dy\,dx \quad (\because x+y+z=1 \Rightarrow z=1-x-y)$$

$$= \int_{x=0}^{1}\int_{y=0}^{1-x} 12x^2y - 3xy + 3y^2 + 3y^2 + 2 - 2x - 2y\, dy\,dx$$

$$= \int_{0}^{1}\int_{0}^{1-x} (12x^2y + 3xy + 3y^2 - 5y - 2x + 2)\, dy\,dx$$

$$= \int_0^1 \left(\frac{12x^2y^2}{2} + \frac{3xy^2}{2} + \frac{3y^3}{3} - \frac{5y^2}{2} - 2xy + 2y \right)_0^{1-x} dx$$

$$= \int_0^1 \left(6x^2(1-x)^2 + \frac{3}{2}x(1-x)^2 + (1-x)^3 - \frac{5}{2}(1-x)^2 - 2x(1-x) + 2(1-x)\right)dx$$

$$= \int_0^1 \left(6x^2(1+x^2-2x) + \frac{3}{2}x(1+x^2-2x) + 1^3 + x^3 - 3x + 3x^2 - \frac{5}{2}(1+x^2-2x) - 2x + 2x^2 + 2 - 2x\right) dx$$

$$= \int_0^1 \left(6x^2 + 6x^4 - 12x^3 + \frac{3}{2}x + \frac{3}{2}x^3 - 3x^2 + 1 - x^3 - 3x + 3x^2 - \frac{5}{2} + \frac{5}{2}x^2 + 5x - 4x + 2x^2 + 2\right) dx$$

$$= \left[6\frac{x^3}{3} + \frac{6x^5}{5} - 12\frac{x^4}{4} + \frac{3}{2}\frac{x^2}{2} + \frac{3}{2}\frac{x^4}{4} - 3\frac{x^3}{3} + x - \frac{x^4}{4} - \frac{3x^2}{2} + \frac{3x^3}{3} - \frac{5}{2}x - \frac{5}{2}\frac{x^3}{3} + \frac{5x^2}{2} - 4\frac{x^2}{2} + 2x^3 + 2x \right]_0^1$$

$$= 2 + \frac{6}{5} + \frac{3}{4} - 3 + \frac{3}{8} + 1 - \frac{1}{4} - \frac{3}{2} - \frac{5}{2} - \frac{5}{6} + \frac{5}{2} - 2 + \frac{2}{3} + 2$$

$$= \frac{6}{5} + \frac{3}{4} + \frac{3}{8} + \frac{2}{3} - \frac{1}{4} - \frac{3}{2} - \frac{5}{6}$$

$$= \frac{49}{120}$$

$$\therefore \int_S \bar{F}.\bar{n}\, dS = \frac{49}{120}$$

5.

x^2

5. Evaluate $\int \bar{F}.\bar{n}\,ds$ where $\bar{F} = z\bar{i} + x\bar{j} - 3y^2z\bar{k}$ and S is the surface $x^2+y^2=16$ included in the first octant between $z=0$ and $z=5$

Let $\phi = x^2+y^2-16$

Let $\nabla\phi$ be the normal to the surface

$$\nabla\phi = \bar{i}\frac{\partial\phi}{\partial x} + \bar{j}\frac{\partial\phi}{\partial y} + \bar{k}\frac{\partial\phi}{\partial z}$$

$$\nabla\phi = 2x\bar{i} + 2y\bar{j}$$

$$\text{unit normal } \bar{n} = \frac{2x\bar{i}+2y\bar{j}}{\sqrt{(2x)^2+(2y)^2}}$$

$$= \frac{2x\bar{i}+2y\bar{j}}{\sqrt{4(x^2+y^2)}}$$

$$= \frac{2x\bar{i}+2y\bar{j}}{\sqrt{4(16)}}$$

$$= \frac{2x\bar{i}+2y\bar{j}}{8}$$

$$\bar{n} = \frac{x}{4}\bar{i} + \frac{y}{4}\bar{j}$$

$$\bar{F}.\bar{n} = (z\bar{i} + x\bar{j} - 3y^2z\bar{k})\left(\frac{x}{4}\bar{i} + \frac{y}{4}\bar{j}\right)$$

$$= \frac{xz}{4}\bar{i} + \frac{xy}{4}\bar{j}$$

for circle $x^2+y^2=16$

Let $x = 4\cos\theta$

$y = 4\sin\theta$

$\theta : 0$ to $\pi/2$ in the first octant

$$\bar{F}.\bar{n} = \frac{(4\cos\theta)z + (4\cos\theta)(4\sin\theta)}{4} = z\cos\theta + 4\sin\theta\cos\theta$$

$r = 4$, $ds = r\,d\theta\,dz$

$ds = 4\,d\theta\,dz$

$$\int_S \bar{F}.\bar{n}\,ds = \int_0^5\int_0^{\pi/2} (z\cos\theta + 4\sin\theta\cos\theta)\,4\,d\theta\,dz$$

$$= 4\int_0^5 \left(z\sin\theta + 2\sin^2\theta\right)_0^{\pi/2} dz$$

$$= 4\int_0^5 (z(1) + 2(1) - 0)\,dz$$

$$= 4\int_0^5 (z+2)\,dz$$

$$= 4\left[\frac{z^2}{2} + 2z\right]_0^5$$

$$= 4\left[\frac{25}{2} + 10\right]$$

$$= 4\left(\frac{45}{2}\right)$$

$$= 90$$

$$\therefore \int_S \bar{F}.\bar{n}\,ds = 90$$

6. Evaluate $\int_V \bar{F}\,dv$, given that $\bar{F} = 2xy\bar{i} - x\bar{j} + y^2\bar{k}$ and V is the region bounded by the surfaces $x=0, x=2, y=0, y=6, z=x^2, z=4$

Given $\bar{F} = 2xz\bar{i} - x\bar{j} + y^2\bar{k}$

$dv = dx\,dy\,dz$

$$\int_V \bar{F}\,dv = \bar{i}\iiint_V f_1\,dx\,dy\,dz + \bar{j}\iiint_V f_2\,dx\,dy\,dz + \bar{k}\iiint_V f_3\,dx\,dy\,dz$$

$$= \bar{i}\int_0^2\int_0^6\int_{x^2}^4 (2xz\,dz)\,dy\,dx + \bar{j}\int_0^2\int_0^6\int_{x^2}^4 (-x\,dz)\,dy\,dx + \bar{k}\int_0^2\int_0^6\int_{x^2}^4 (y^2\,dz)\,dy\,dx$$

$$= \bar{i}\int_0^2\int_0^6 \left(2x\frac{z^2}{2}\right)_{x^2}^4 dy\,dx + \bar{j}\int_0^2\int_0^6 -x(z)_{x^2}^4\,dy\,dx + \bar{k}\int_0^2\int_0^6 y^2(z)_{x^2}^4\,dy\,dx$$

$$= \bar{i}\int_0^2\int_0^6 x(16-x^4)\,dy\,dx + \bar{j}\int_0^2\int_0^6 -x(4-x^2)\,dy\,dx + \bar{k}\int_0^2\int_0^6 y^2(4-x^2)\,dy\,dx$$

$$= \bar{i}\int_0^2 \left[16xy - x^5y\right]_0^6 dx + \bar{j}\int_0^2 \left[4xy - x^3y\right]_0^6 dx + \bar{k}\int_0^2 \left[\frac{4y^3}{3} - x^2\frac{y^3}{3}\right]_0^6 dx$$

$$= \bar{i}\int_0^2 (96x - 6x^5)\,dx - \bar{j}\int_0^2 (24x - 6x^3)\,dx + \bar{k}\int_0^2 (288 - x^2 72)\,dx$$

$$= \bar{i}\left(\frac{96x^2}{2} - \frac{6x^6}{6}\right)_0^2 - \bar{j}\left(\frac{24x^2}{2} - \frac{6x^4}{4}\right)_0^2 + \bar{k}\left(288x - \frac{24}{72}\frac{x^3}{3}\right)_0^2$$

$$= \bar{i}\,(48(4) - 64) - \bar{j}\,(12(4) - 24) + \bar{k}\,(288(2) - 24(8))$$

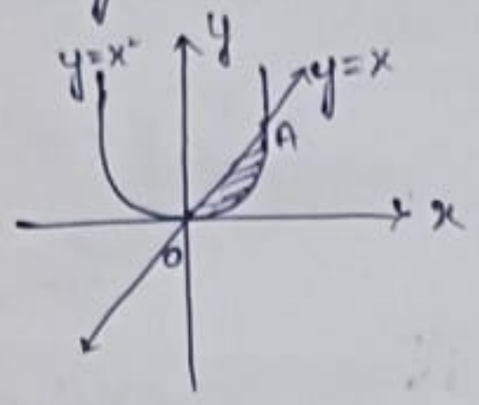
$$= 128\bar{i} - 24\bar{j} + 8\bar{k}(576 - 192)$$

$\therefore \int_V \vec{F}\,dv = 128\vec{i} - 24\vec{j} + 384\vec{k}$

7. Verify Green's theorem for $\int_C [(xy+y^2)dx + x^2dy]$, where C is bounded by $y=x$ and $y=x^2$

Given $\int_C [(xy+y^2)dx + x^2dy]$

$y=x$, $y=x^2$



By Green's theorem

$$\int_C (M\,dx + N\,dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy \quad \text{—①}$$

On OA: $y = x^2 \Rightarrow dy = 2x\,dx$

$$(xy+y^2)dx + x^2dy = (x(x^2)+(x^2)^2)\,dx + (2x\,dx)x^2$$

$$= (x^3 + x^4 + 2x^3)\,dx$$

$$= (3x^3 + x^4)\,dx$$

$x: 0 \text{ to } 1$

$$\int_{OA} (xy+y^2)dx + x^2dy = \int_0^1 (x^4 + 3x^3)\,dx$$

$$= \left[\frac{x^5}{5} + \frac{3x^4}{4}\right]_0^1$$

$$= \frac{1}{5} + \frac{3}{4}$$

$$= \frac{19}{20}$$

Along AO:

$y = x \quad dy = dx$

$$M\,dx + N\,dy = (xy+y^2)\,dx + x^2\,dy$$

$$= (x(x)+x^2)\,dx + x^2\,dx$$

$$= 3x^2\,dx$$

$x: 1 \text{ to } 0$

$$\int_{AO}(xy+y^2)dx + x^2dy = \int_0^0 3x^2dx$$

$$= \left[\frac{3x^3}{3}\right]_0^0$$

$$= 0 - (1)$$

$$= -1$$

$$\int_C (M\,dx + N\,dy) = \int_{OA}(xy+y^2)dx + x^2dy + \int_{AO}(xy+y^2)dx + x^2dy$$

$$= \frac{19}{20} - 1$$

$$= \frac{-1}{20}$$

$M = xy + y^2 \qquad N = x^2$

$\frac{\partial M}{\partial y} = x + 2y \qquad \frac{\partial N}{\partial x} = 2x$

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 2x - x - 2y$$

$$= x - 2y$$

Fix x

x : 0 to 1

y : x^2 to x

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)dx\,dy = \int_0^1 \int_{x^2}^{x} (x-2y)\,dy\,dx$$

$$= \int_0^1 \left[x(y)_{x^2}^{x} - 2\left(\frac{y^2}{2}\right)_{x^2}^{x}\right]dx$$

$$= \int_0^1 \left[x(x-x^2) - (x^2 - (x^2)^2)\right]dx$$

$$= \int_0^1 (x^2 - x^3 - x^2 + x^4)\,dx = \int_0^1 x^4 - x^3\,dx$$

$$= \left(\frac{x^5}{5} - \frac{x^4}{4}\right)_0^1$$

$$= \frac{1}{5} - \frac{1}{4} = -1/20$$

$$\therefore \int_C M\,dx + N\,dy = \iint_R \left(\frac{\partial N}{\partial x} + \frac{\partial M}{\partial y}\right)dx\,dy$$

Hence, Green's theorem is verified

8. Evaluate by Green's theorem $\int_C [(y-\sin x)\,dx + \cos x\,dy]$, where C is the triangle enclosed with the lines $y=0,\ x=\pi/2,\ y=\frac{2}{\pi}x$

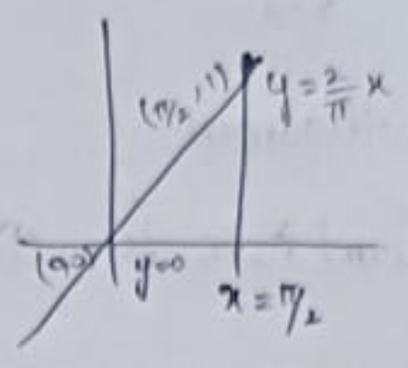
Given $\int_C (y-\sin x)\,dx + \cos x\,dy$

Let $M = (y-\sin x)$ $\quad N = \cos x$

$$\frac{\partial M}{\partial y} = 1 \qquad \frac{\partial N}{\partial x} = \sin x$$

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = -\sin x - 1 = -(1+\sin x)$$

Given $y=0,\ x=\pi/2,\ y=\frac{\pi}{2}x$



Fix x:

$x = 0$ to $\pi/2$

$y = 0$ to $\frac{2}{\pi}x$

By Green's theorem,

$$\int_C (M\,dx + N\,dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$$

$$\int_C (y-\sin x)\,dx + \cos x\,dy = \int_0^{\pi/2}\int_0^{2/\pi x} (1+\sin x)\,dy\,dx$$

$$= -\int_0^{\pi/2} (1+\sin x)\left[\int_{y=0}^{2/\pi x} dy\right] dx$$

$$= -\int_{x=0}^{\pi/2} (1+\sin x)\,(y)_0^{2/\pi x}\,dx$$

$$= -\int_{x=0}^{\pi/2} (1+\sin x)\,\frac{2x}{\pi}\,dx$$

$$= \frac{-2}{\pi}\int_0^{\pi/2} (1+\sin x)\,x\,dx$$

$$= \frac{-2}{\pi}\int_0^{\pi/2} x\,dx + \int_{x=0}^{\pi/2} x\sin x\,dx$$

$$= \frac{-2}{\pi}\int_0^{\pi/2} x^1\,dx + \left[x(-\cos x) - (1)(-\sin x)\right]_0^{\pi/2}$$

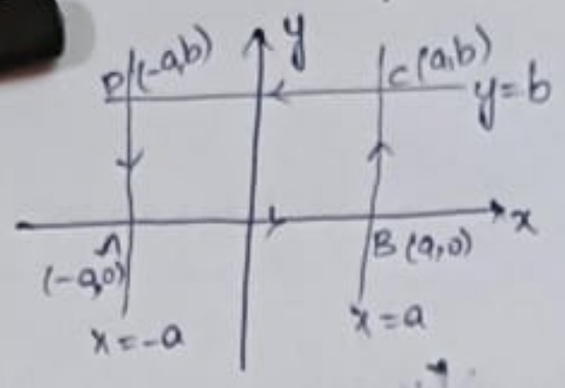
$$= -\frac{2}{\pi}\left(\frac{x^2}{2}\right)_0^{\pi/2} + \left(0+1-(0)\right)$$

$$= \frac{-2}{\pi}\left(\frac{\pi^2}{8}+1\right)$$

$$= \frac{-\pi}{4} + \frac{2}{\pi}$$

9. Verify stoke's theorem for $\bar{F} = (x^2+y^2)\bar{i} - 2xy\bar{j}$ taken around the rectangle bounded by the lines $x = \pm a, y = 0, y = b$

Given $x = \pm a, y = 0, y = b$



By Stoke's theorem

$$\int_C \bar{F}\cdot d\bar{r} = \int_S \text{curl}\,\bar{F}\cdot\bar{n}\, ds \quad -①$$

$$\int_C \bar{F}\cdot d\bar{r} = \int_{AB} \bar{F}\cdot d\bar{r} + \int_{BC} \bar{F}\cdot d\bar{r} + \int_{CD} \bar{F}\cdot d\bar{r} + \int_{DA} \bar{F}\cdot d\bar{r} \quad -②$$

Given

$$\bar{F} = (x^2+y^2)\bar{i} - 2xy\bar{j}$$

Let $\bar{r} = x\bar{i} + y\bar{j}$

$$d\bar{r} = \bar{i}dx + \bar{j}dy$$

$$\bar{F}\cdot d\bar{r} = (x^2+y^2)dx - 2xy\,dy$$

Along AB

$y = 0 \Rightarrow dy = 0$

$$\bar{F}\cdot d\bar{r} = (x^2+0^2)dx + 0$$

$$= x^2 dx$$

$x: -a$ to a

$$\int_{AB} \bar{F}\cdot d\bar{r} = \int_{x=-a}^{a} x^2 dx$$

$$= \left[\frac{x^3}{3}\right]_{-a}^{a}$$

$$= \frac{1}{3}(a^3+a^3)$$

$$= \frac{2a^3}{3}$$

Along BC

$x = a \Rightarrow dx = 0$

$$\bar{F}\cdot d\bar{r} = -2ay\,dy$$

$y: 0$ to b

$$\int_{BC} \bar{F}\cdot d\bar{r} = \int_{y=0}^{b} -2ay\,dy$$

$$= \left[\frac{-2ay^2}{2}\right]_0^b$$

$$= (-ay^2)_0^b$$

$$= -ab^2 - 0$$

$$= -ab^2$$

Along CD

$y = b, \; dy = 0$

$$\bar{F}\cdot d\bar{r} = (x^2+b^2)dx$$

$x: a$ to $-a$

$$\int_{CD} \bar{F}\cdot d\bar{r} = \int_{x=a}^{-a} (x^2+b^2)dx$$

$$= \left(\frac{x^3}{3} + b^2x\right)_a^{-a}$$

$$= \frac{-a^3}{3} + b^2(-a) - \frac{a^3}{3} - ab^2$$

$$= -\frac{a^3}{3} - ab^2 - \frac{a^3}{3} - ab^2$$

$$= -2\left(\frac{a^3}{3} + ab^2\right)$$

Along DA

$x = -a \Rightarrow dx = 0$

$$\bar{F}\cdot d\bar{r} = -2(-a)\,y\,dy$$

$$= 2ay\,dy$$

$$\int \bar{F}\cdot d\bar{r} = \int_b^0 2ay\,dy$$

$$= \left(\frac{2ay^2}{2}\right)_b^0$$

$= (ay^2)_b^0$

$= a(0) - a(b^2)$

$= -ab^2$

from ②

$$\int_C \bar{F}.d\bar{r} = \frac{2}{3}a^3 - ab^2 - \frac{2a^3}{3} - 2ab^2 - ab^2$$

$$= -4ab^2$$

$$\text{Curl } \bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2+y^2 & -2xy & a \end{vmatrix}$$

$$= \bar{i}\left(\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(-2xy)\right) - \bar{j}\left(\frac{\partial}{\partial x}(0) - \frac{\partial}{\partial z}(x^2+y^2)\right) + \bar{k}\left(\frac{\partial}{\partial x}(-2xy) - \frac{\partial}{\partial y}(x^2+y^2)\right)$$

$= \bar{k}(-2y - 2y)$

$= \bar{k}(-4y)$

$= -4y\bar{k}$

$\text{curl } \bar{F} = -4y\bar{k}$

$\bar{n} = \bar{k}$

$\text{Curl } \bar{F}.\bar{n} = -4y.$

$ds = dx\,dy$

$x: -a \text{ to } +a$

$y: 0 \text{ to } b$

$$\int \text{curl } \bar{F}.\bar{n}\,ds = \int_{x=-a}^{a}\int_{0}^{b} -4y\,dy\,dx$$

$$= -4\int_{-a}^{a}\left(\frac{y^2}{2}\right)_0^b dx$$

$$= -4\int_{-a}^{a}\frac{b^2}{2}dx$$

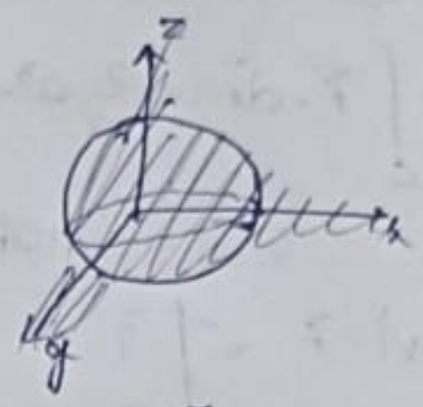
$$= \frac{-4b^2}{2}\int_{-a}^{a} dx$$

$$= -\frac{4b^2}{2}(x)_{-a}^{a} = -4ab^2$$

$\therefore \int_C \bar{F}.d\bar{r} = \int_S \text{curl } \bar{F}.\bar{n}\,ds$ Hence stoke's theorem verified

10. Verify Stoke's theorem for the vector field $\bar{F} = (2x-y)\bar{i} - yz^2\bar{j} - y^2z\bar{k}$ over the upper half of the sphere $x^2+y^2+z^2=1$ bounded by its projection on the x-y plane.

Given $\bar{F} = (2x-y)\bar{i} - yz^2\bar{j} - y^2z\bar{k}$

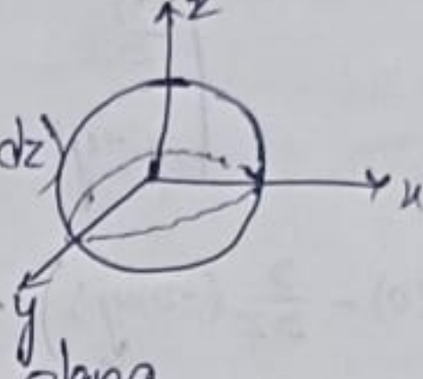


By stoke's theorem

$$\int_C \bar{F}.d\bar{r} = \int_S \text{curl}\,\bar{F}.\bar{n}\,ds \longleftarrow (1)$$

Let $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$

$$d\bar{r} = \bar{i}\,dx + \bar{j}\,dy + \bar{k}\,dz$$



$$\bar{F}.d\bar{r} = (2x-y)\bar{i} - yz^2\bar{j} - y^2z\bar{k})(\bar{i}\,dx + \bar{j}\,dy + \bar{k}\,dz)$$

$$\bar{F}.d\bar{r} = (2x-y)\,dx - yz^2\,dy - y^2z\,dz$$

the sphere $x^2+y^2+z^2=1$ projection on the x-y plane

$$z=0,\quad dz=0$$

$$\int_C \bar{F}.d\bar{r} = \int_C (2x-y)\,dx - yz^2\,dy - y^2z\,dz$$

$$\int_C \bar{F}.d\bar{r} = \int_C (2x-y)\,dx$$

Let $x = \cos\theta \quad dx = -\sin\theta\,d\theta$

$y = \sin\theta$

$\theta: 0$ to 2π

$$= \int_0^{2\pi} (2\cos\theta - \sin\theta)(-\sin\theta\,d\theta)$$

$$= \int_0^{2\pi} (-2\sin\theta\cos\theta + \sin^2\theta)\,d\theta$$

$$= -\int_0^{2\pi} \sin^2 2\theta\,d\theta + \int_0^{2\pi} \sin^2\theta\,d\theta$$

$$= \left(\frac{\cos 2\theta}{2}\right)_0^{2\pi} + 4\int_0^{\pi/2} \sin^2\theta\,d\theta$$

$$= \left(\frac{1}{2} - \frac{1}{2}\right) + 4\cdot\frac{(2-1)}{2}\cdot\frac{\pi}{2}$$

$$\therefore \int_C \bar{F}.d\bar{r} = \pi$$

Now $\text{curl}\,\bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x-y & -yz^2 & -y^2z \end{vmatrix}$

$= \bar{i}\left(\frac{\partial}{\partial y}(-y^2z) - \frac{\partial}{\partial z}(-yz^2)\right) - \bar{j}\left(\frac{\partial}{\partial x}(-y^2z) - \frac{\partial}{\partial z}(2x-y)\right) + \bar{k}\left(\frac{\partial}{\partial x}(-yz^2) - \frac{\partial}{\partial y}(2x-y)\right)$

$= \bar{i}(-2yz + 2yz) - \bar{j}(0-0) + \bar{k}(0-(-1))$

$= \bar{i}(0) + \bar{j}(0) + \bar{k}$

$= \bar{k}$

$\text{curl}\,\bar{F} = \bar{k}$

$\bar{n} = x\bar{i} + y\bar{j} + z\bar{k}$

$\text{curl}\,\bar{F} \cdot \bar{n} = \bar{k} \cdot (x\bar{i} + y\bar{j} + z\bar{k})$

$= z$

from ①

$\int_S \text{curl}\,\bar{F} \cdot \bar{n}\, ds = \int_S z\, ds$

$ds = \frac{dA}{|\bar{n} \cdot k|} = \frac{dA}{z}$

$= \int_S z\left(\frac{1}{z}\right) dA$

$= \int_S dA = \int_R dA$

$\therefore \int_R dA$ is the area of the projection R

since R is circle with radius r=1

$\text{Area} = \pi r^2 = \pi(1)^2 = \pi$

$\therefore \int_S \text{curl}\,\bar{F} \cdot \bar{n}\, ds = \pi$

$\therefore \int_C \bar{F} \cdot d\bar{r} = \int_S \text{curl}\,\bar{F} \cdot \bar{n}\, ds$

Hence stoke's theorem verified.

11. Verify Gauss Divergence theorem for $\bar{F}=(x^2-yz)\bar{i}+(y^2-zx)\bar{j}+(z^2-xy)\bar{k}$ taken over the rectangular parallelepiped $0\le x\le a, 0\le y\le b, 0\le z\le c$.

Given $\bar{F}=(x^2-yz)\bar{i}+(y^2-zx)\bar{j}+(z^2-xy)\bar{k}$

$$\text{div}\,\bar{F}=\frac{\partial}{\partial x}(x^2-yz)+\frac{\partial}{\partial y}(y^2-zx)+\frac{\partial}{\partial z}(z^2-xy)$$

$$=2x+2y+2z$$

$$=2(x+y+z)$$

$$dv=dx\cdot dy\cdot dz$$

$x: 0$ to a

$y: 0$ to b

$z: 0$ to c

By divergence theorem

$$\int_S \bar{F}\cdot\bar{n}\,ds=\int_V \text{div}\,\bar{F}\,dv \quad -(1)$$

from (1)

$$\int_V \text{div}\,\bar{F}\,dv=2\int_0^a\int_0^b\int_0^c (x+y+z)\,dx\,dy\,dz$$

$$=2\int_0^a\int_0^b\left[(x+y+z)(z)\right]_0^c dy\,dx$$

$$=2\int_0^a\int_0^b\left(xz+yz+\frac{z^2}{2}\right)_0^c dy\,dx$$

$$=2\int_0^a\int_0^b\left(xc+yc+\frac{c^2}{2}\right)dy\,dx$$

$$=2\int_0^a\left(xcy+\frac{cy^2}{2}+\frac{c^2y}{2}\right)_0^b dx$$

$$=2\int_0^a\left(xcb+\frac{cb^2}{2}+\frac{c^2b}{2}\right)dx$$

$$=2\left[\frac{x^2cb}{2}+\frac{cb^2x}{2}+\frac{c^2bx}{2}\right]_0^a$$

$$=\frac{2}{2}\left[a^2cb+acb^2+c^2ba\right]$$

$$=abc(a+b+c)$$

$$\therefore \int_V \text{div}\,\bar{F}\,dv=abc(a+b+c)$$

Now $\int_S \bar{F}\cdot\bar{n}\,ds = \int_{S_1} \bar{F}\cdot\bar{n}\,ds + \int_{S_2} \bar{F}\cdot\bar{n}\,ds + \int_{S_3} \bar{F}\cdot\bar{n}\,ds + \int_{S_4} \bar{F}\cdot\bar{n}\,ds + \int_{S_5} \bar{F}\cdot\bar{n}\,ds + \int_{S_6} \bar{F}\cdot\bar{n}\,ds$ ——— (2)

At S_1 : $x=0$

$dx=0$

$\bar{n}=-\bar{i}$ $\quad ds = dydz$

$y: 0$ to b

$z: 0$ to c

$\bar{F}\cdot\bar{n} = -(x^2-yz)$

$$\int_{S_1} \bar{F}\cdot\bar{n}\,ds = \int_0^b\int_0^c -(x^2-yz)\,dz\,dy$$

$$= \int_0^b\int_0^c (yz)\,dy\,dz = \int_0^b \left(y\frac{z^2}{2}\right)_0^c dz$$

$$= \int_0^b \left(y\frac{c^2}{2}\right) dz$$

$$= \left(\frac{y^2c^2}{4}\right)_0^b$$

$$\int_{S_1} \bar{F}\cdot\bar{n}\,ds = \frac{b^2c^2}{4}$$

At S_2 : $x=a$

$dx=0$

$\bar{n}=\bar{i}$ $\quad ds = dydz$

$y: 0$ to b

$z: 0$ to c

$\bar{F}\cdot\bar{n} = x^2+yz$

$$\int_{S_2} \bar{F}\cdot\bar{n}\,ds = \int_0^b\int_0^c (x^2-yz)\,dz\,dy$$

$$= \int_0^b \left(a^2z - y\frac{z^2}{2}\right)_0^c dy$$

$$= \int_0^b \left(a^2c - y\frac{c^2}{2}\right) dy$$

$$= \left(a^2cy - \frac{y^2c^2}{4}\right)_0^b$$

$$\int_{S_2} \bar{F}\cdot\bar{n}\,ds = a^2cb - \frac{b^2c^2}{4}$$

At S_3 : $y=0 \Rightarrow dy=0$

$\bar{n}=-\bar{j}$ $\quad ds = dxdz$

$x: 0$ to a

$z: 0$ to c

$\bar{F}\cdot\bar{n} = -(y^2-zx)$

$$\int_{S_3} \bar{F}\cdot\bar{n}\,ds = \int -(y^2-zx)\,dx\,dz$$

$$= \int_0^a\int_0^c zx\,dx\,dz$$

$$= \int_0^a x\left(\frac{z^2}{2}\right)_0^c dx$$

$$= \int_0^a \left(x\frac{c^2}{2}\right) dx$$

$$= \left(\frac{x^2}{2}\frac{c^2}{2}\right)_0^a$$

$$= \frac{a^2b^2}{4}$$

At S_4 : $y=b \Rightarrow dy=0$

$\bar{n}=\bar{j}$ $\quad ds = dxdz$

$x: 0$ to a

$z: 0$ to c

$\bar{F}\cdot\bar{n} = +(y^2-zx)$

$$\int_{S_4} \bar{F}\cdot\bar{n}\,ds = \int (y^2-zx)\,dx\,dz$$

$$= \int_0^a\int_0^c (y^2-zx)\,dx\,dz$$

$$= \int_0^a\int_0^c (b^2-zx)\,dx\,dz$$

$$= \int_0^a \left(b^2z - x\frac{z^2}{2}\right)_0^c dx$$

$$= \int_0^a \left(b^2c - x\frac{c^2}{2}\right) dx = \left(b^2cx - \frac{x^2c^2}{4}\right)_0^a$$

$$= acb^2 - \frac{b^2c^2}{4}$$

At S_5: $z=0$ $dz=0$

$\bar{n}=-\bar{k}$ $ds=dx\,dy$

$x: 0$ to a

$y: 0$ to b

$\bar{F}.\bar{n} = -(z^2-xy)$

$$\int_{S_5} \bar{F}.\bar{n}\,ds = \int_0^a\int_0^b (xy)\,dx\,dy$$

$$= \int_0^a \left(\frac{xy^2}{2}\right)_0^b dx$$

$$= \int_0^a \frac{xb^2}{2}\,dx$$

$$= \left(\frac{x^2b^2}{2\cdot 2}\right)_0^a$$

$$= \frac{a^2b^2}{4}$$

At S_6: $z=c$ $dz=0$

$\bar{n}=\bar{k}$ $ds=dx\,dy$

$x: 0$ to a

$y: 0$ to b

$\bar{F}.\bar{n} = z^2-xy$

$$\int_{S_6} \bar{F}.\bar{n}\,ds = \int_0^a\int_0^b (c^2-xy)\,dx\,dy$$

$$= \int_0^a \left(c^2y - \frac{xy^2}{2}\right)_0^b dx$$

$$= \int_0^a \left(c^2b - x\frac{b^2}{2}\right) dx$$

$$= \left(c^2bx - \frac{x^2}{2}\frac{b^2}{2}\right)_0^a$$

$$\int_{S_6} \bar{F}.\bar{n}\,ds = \left(abc^2 - \frac{a^2b^2}{4}\right)$$

from eq ②

$$\int_S \bar{F}.\bar{n}\,ds = \int_{S_1} \bar{F}.\bar{n}\,ds + \int_{S_2} \bar{F}.\bar{n}\,ds + \int_{S_3} \bar{F}.\bar{n}\,ds + \int_{S_4} \bar{F}.\bar{n}\,ds + \int_{S_5} \bar{F}.\bar{n}\,ds + \int_{S_6} \bar{F}.\bar{n}$$

$$= \frac{b^2c^2}{4} + a^2cb - \frac{b^2c^2}{4} + \frac{a^2c^2}{4} + ab^2c - \frac{a^2c^2}{4} + \frac{a^2b^2}{4} + abc^2 - \frac{a^2b^2}{4}$$

$$= a^2cb + ab^2c + abc^2$$

$$\int_S \bar{F}.\bar{n}\,ds = abc(a+b+c)$$

$$\therefore \int_S \bar{F}.\bar{n}\,ds = \int_V \text{div}\,\bar{F}\,dv$$

Hence Gauss Divergence theorem verified.

12. Evaluate $\int_S [x\,dy\,dx + y\,dz\,dx + z\,dx\,dy]$ using divergence theorem where S is the surface the sphere $x^2+y^2+z^2=a^2$

Let $\bar{F} = x\bar{i} + y\bar{j} + z\bar{k}$

$$\text{div}\,\bar{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z)$$

$$= 1+1+1$$

$$= 3$$

By Gaus Divergence theorem

$$\int_S \bar{F}.\bar{n}\,ds = \int_V \text{div}\,\bar{F}\,dV$$

$$= \int_V 3\,dV$$

$$= 3[V]$$

$$= 3.\frac{4}{3}\pi(a)^3$$

$$\int_S \bar{F}.\bar{n}\,ds = 4a^3\pi$$

13. State the following vector integral theorems:

i) Green's Theorem: (transformation between line and double integrals)

If R is a closed region in x-y plane bounded by a simple closed curve C and if M and N are continously differentiable functions of x and y in R then

$$\int_C (M\,dx + N\,dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy.$$ where C is traversed in the positive direction.

ii) Stoke's theorem: (Transformation between line and surface integral)

Let S be an open surface bounded by a closed non intersecting curve C if $\bar{F}$ is any vector differentiable function then

$$\int_C \bar{F}.d\bar{r} = \int_S \text{curl}\,\bar{F}.\bar{n}\,ds$$, where C is traversed in positive direction and $\bar{n}$ is outward down unit normal at any point of the surface.

Gauss-divergence theorem: (Transformation btw surface and volume integrals)

Let S be a closed surface enclosing a volume ρ. If $\bar{F}$ is a continously differentiable vector point function then $\int_S \bar{F}.\bar{n}\,ds = \int_v \text{div}\,\bar{F}\,dv$, where $\bar{n}$ is the outward drawn unit normal vector at any point of s